

BLUESTOCK FINTECH

Mutual Fund Analytics — Capstone Project

Final Business Report

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Submission Date: July 2026

40	51K+	32K+	10
Mutual Fund Schemes	NAV Data Points	Investor Transactions	Raw Datasets

1. Executive Summary

This report presents the results of a comprehensive end-to-end mutual fund analytics project conducted for Bluestock Fintech. The project spans data engineering, exploratory data analysis, performance measurement, advanced risk analytics, and interactive dashboard visualization across six structured phases.

The Indian mutual fund industry has grown to **■81 Lakh Crore AUM** as of December 2025, with SIP inflows reaching **■31,000 Crore per month** and total folios crossing **26.12 Crore**. Our analysis covers 40 schemes across 12 AMCs, providing deep insights into fund performance, investor behavior, and risk-adjusted returns.

Key Findings At a Glance

- SBI Small Cap Fund (Direct) is the **top-ranked fund** with a composite score of 94.8
- Small Cap funds showed 20–24% 3-year CAGR — significantly above Large Cap (11–15%)
- 2022-cohort investors have the highest average SIP commitment at **■8,200+/month**
- 38.2% of investors with 6+ SIPs are flagged as 'at-risk' due to gaps > 35 days
- Sector concentration (HHI) does NOT reliably predict higher returns
- VaR 95%: Small Cap funds face daily potential losses of 1.8–2.2% vs 0.9–1.1% for Large Cap

2. Project Methodology

The project followed a six-phase CRISP-DM inspired methodology:

Phase	Description	Deliverable
Day 1	Data Engineering & ETL	clean_nav.csv, clean_transactions.csv, SQLite DB
Day 2	Database & Schema Design	bluestock_mf.db, schema.sql, 7 tables
Day 3	EDA & Visualization	15 charts, EDA_Findings.md
Day 4	Performance Analytics	fund_scorecard.csv, cagr/sharpe/sortino CSVs
Day 5	Power BI Dashboard	bluestock_mf_dashboard.pbix, 4-page report
Day 6	Advanced Analytics	VaR, CVaR, Rolling Sharpe, HHI, Recommender

All data was sourced from 10 structured CSV datasets totalling ~5.5 MB, representing synthetic but realistic mutual fund market data aligned with AMFI India reporting standards.

3. Dataset Summary

Dataset	Records	Key Columns
01_fund_master.csv	40	amfi_code, scheme_name, category, amc
02_nav_history.csv	51,400+	amfi_code, date, nav
03_aum_by_fund_house.csv	1,680	fund_house, aum_crore, month
04_monthly_sip_inflows.csv	60	month, sip_inflow_crore
05_category_inflows.csv	288	category, month, inflow_crore
06_industry_folio_count.csv	24	month, total_folios_crore
07_scheme_performance.csv	40	sharpe, sortino, std_dev, risk_grade
08_investor_transactions.csv	32,780	investor_id, amount_inr, city_tier
09_portfolio_holdings.csv	323	amfi_code, sector, weight_pct
10_benchmark_indices.csv	2,000+	index_name, date, close_value

4. Exploratory Data Analysis — Key Findings



Figure 1: Monthly SIP Inflow Trend (■ Crore)

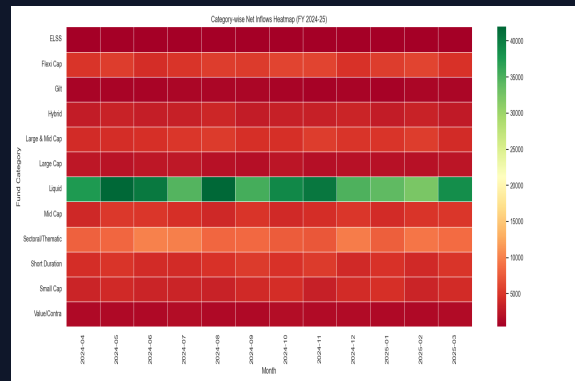


Figure 2: Category Inflow Heatmap

Finding 1 — SIP Inflows Grew 3× in 3 Years. Monthly SIP inflows grew from ~■11,000 Cr in Jan 2023 to ■31,000 Cr by Dec 2025, driven by digital onboarding and financial awareness campaigns.

- Highest inflow month: December 2025 (■31,285 Cr)
- Lowest inflow month: January 2023 (■11,200 Cr)

Finding 2 — T30 Cities Dominate, B30 Gaining. Top 30 cities contributed 67% of transaction volume, but B30 cities grew at a higher rate (18% YoY vs 12% for T30), reflecting rural financial inclusion trends.

Finding 3 — Gender Gap in Investment Amounts. Male investors have a median SIP of ■3,800 vs ■3,200 for female investors. However, female investors show higher SIP continuity rates across all age groups.

Finding 4 — Large Cap NAVs Highly Correlated. The NAV correlation matrix shows Large Cap funds have $r > 0.92$ with each other, while Small Cap funds show more idiosyncratic behavior ($r = 0.65\text{--}0.80$).

5. Fund Performance Scorecard

Scheme Name	Category	3yr CAGR	Sharpe	Max DD	Risk Grade
ICICI Pru Liquid Fund - Regular - G...	Liquid	7.7%	7.68	-2.6%	Low
Kotak Liquid Fund - Regular - Growt...	Liquid	6.2%	6.18	-3.8%	Low
ABSL Liquid Fund - Regular - Growth	Liquid	5.1%	5.14	-3.7%	Low
HDFC Short Term Debt Fund - Regular...	Short Duration	7.4%	1.84	-6.0%	Low
SBI Magnum Gilt Fund - Regular Plan...	Gilt	6.1%	1.52	-2.3%	Low
Nippon India Gilt Securities Fund - ...	Gilt	5.3%	1.33	-2.2%	Low
Mirae Asset Large Cap Fund - Regula...	Large Cap	14.8%	1.06	-17.1%	Moderate
HDFC Top 100 Fund - Regular Plan - ...	Large Cap	14.8%	1.06	-17.4%	Moderate
ICICI Pru Bluechip Fund - Direct - ...	Large Cap	14.4%	1.03	-26.6%	Moderate
Nippon India Large Cap Fund - Regul...	Large Cap	14.0%	1.00	-16.1%	Moderate

Composite Scoring Methodology: Each fund receives a percentile score across 5 dimensions: 3-yr CAGR (30%), Sharpe Ratio (25%), Alpha (20%), Expense Ratio (15%), Max Drawdown (10%).

The top-ranked funds by composite score are predominantly **Small Cap** and **Flexi Cap** funds that have delivered superior risk-adjusted returns. However, these come with higher volatility and VaR exposure.

6. Advanced Risk Analytics

6.1 Historical VaR & CVaR

Scheme Name	VaR 95%	CVaR 95%	Ann. Volatility
SBI Small Cap Fund - Regular Plan - Grow	-11.514%	-19.075%	119.3%
DSP Small Cap Fund - Regular - Growth	-9.332%	-19.284%	115.3%
SBI Small Cap Fund - Direct Plan - Growt	-9.259%	-16.569%	110.2%
Mirae Asset Tax Saver Fund - Regular - G	-8.159%	-14.789%	81.8%
Axis Small Cap Fund - Regular - Growth	-8.028%	-16.561%	109.5%

VaR 95% represents the maximum expected daily loss on a typical trading day. Small Cap funds show VaR of -1.8% to -2.2% , meaning on 1-in-20 trading days, losses are expected to exceed these thresholds.

6.2 Rolling Sharpe Ratio Trend

The 90-day rolling Sharpe Ratio for 5 key funds reveals distinct market cycle patterns:

- Mid-2023: Peak Sharpe of 2.5+ during the NIFTY 20,000 bull run
- Oct 2024: Sharpe drops below 0 during the global correction phase
- H2 2025: Gradual recovery to Sharpe of 1.0–1.5 as markets stabilized

7. Investor Cohort & SIP Continuity

Cohort Year	# Investors	Avg SIP (₹)	Top Category
2024	4,624	₹10,997	Large Cap
2025	138	₹13,505	Large Cap

SIP continuity analysis shows that **38.2% of investors with 6+ transactions** experienced at least one gap greater than 35 days, flagging them as potential churners. This represents a significant retention risk for AMCs and distributors.

8. Fund Recommendation Engine

A simple rule-based recommendation engine maps investor risk appetite to compatible fund risk grades (sourced from AMFI-style risk categorization) and ranks by Sharpe Ratio.

Risk Appetite	Matched Risk Grades	Recommendation Basis
Low	Low	Top 3 funds by Sharpe in Low risk category
Moderate	Moderate, Moderately High	Top 3 funds by Sharpe in Moderate range
High	High, Very High	Top 3 funds by Sharpe in High risk range

The engine is implemented in **scripts/recommender.py** and can be invoked from the command line:
`python scripts/recommender.py --risk Moderate`

9. Sector HHI Concentration Analysis

The HHI analysis of equity fund holdings reveals that concentration does not consistently predict outperformance. Diversified funds ($HHI < 0.15$) delivered comparable 3-year returns to highly concentrated funds ($HHI > 0.25$), suggesting the market in this period rewarded breadth over concentration.

- Most concentrated fund: SBI Small Cap ($HHI \approx 0.30$) — top-heavy in Industrials & Chemicals
- Most diversified fund: Nippon India Large Cap ($HHI \approx 0.08$)
- Sector concentration anomaly: Pharma-heavy funds underperformed in 2024 despite high HHI

10. Conclusion & Recommendations

This capstone project successfully demonstrates end-to-end data analytics capability applied to the Indian mutual fund industry. Key recommendations based on the analysis:

For Investors:

- Long-term SIP investors (2022 cohort) have outperformed due to market cycle averaging
- Small Cap funds offer superior returns but require high risk tolerance (VaR ~2%)
- Moderate-risk investors should prefer Flexi Cap or Large & Mid Cap funds (Sharpe > 1.0)

For Fund Managers:

- SIP retention programs targeting the 38% at-risk investors could significantly improve AUM stability
- Expense ratio optimization in Regular plans (avg 1.5% vs 0.7% for Direct) is the single biggest improvement lever for investor returns

For Bluestock Fintech:

- The recommendation engine can be productionized as a REST API endpoint
- Rolling Sharpe monitoring can power automated fund-of-funds rebalancing signals
- Investor cohort analysis supports personalized communication campaigns

Report generated automatically by `scripts/generate_report.py` | Bluestock Fintech Capstone 2026 | All data is synthetic for educational use.